

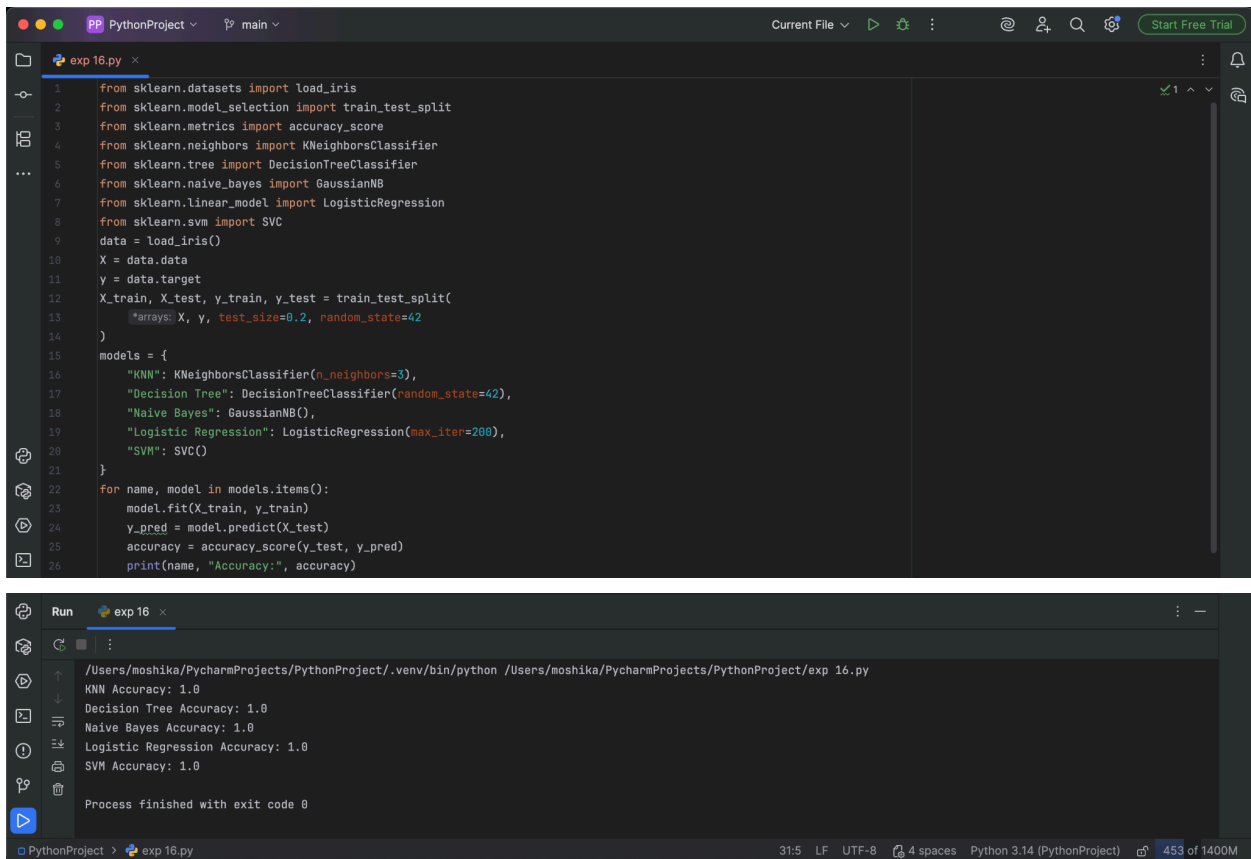
EXPERIMENT - 16

Aim: To compare different classification algorithms and evaluate their performance.

Procedure:

1. Load a suitable classification dataset.
2. Preprocess the dataset.
3. Split the data into training and testing sets.
4. Apply different classification algorithms such as KNN, Decision Tree, Naive Bayes, and Logistic Regression.
5. Train each model using the training data.
6. Predict the classes for test data.
7. Calculate the accuracy of each algorithm.
8. Compare the obtained performance.
9. Display the results.

Output:



The image displays a Python IDE with a code editor and a run console. The code in the editor imports necessary libraries (sklearn), loads the Iris dataset, splits it into training and testing sets, and then trains and evaluates four models: KNeighborsClassifier, DecisionTreeClassifier, GaussianNB, and LogisticRegression. The output in the console shows that all four models achieved an accuracy of 1.0.

```
1 from sklearn.datasets import load_iris
2 from sklearn.model_selection import train_test_split
3 from sklearn.metrics import accuracy_score
4 from sklearn.neighbors import KNeighborsClassifier
5 from sklearn.tree import DecisionTreeClassifier
6 from sklearn.naive_bayes import GaussianNB
7 from sklearn.linear_model import LogisticRegression
8 from sklearn.svm import SVC
9 data = load_iris()
10 X = data.data
11 y = data.target
12 X_train, X_test, y_train, y_test = train_test_split(
13     X, y, test_size=0.2, random_state=42
14 )
15 models = {
16     "KNN": KNeighborsClassifier(n_neighbors=3),
17     "Decision Tree": DecisionTreeClassifier(random_state=42),
18     "Naive Bayes": GaussianNB(),
19     "Logistic Regression": LogisticRegression(max_iter=200),
20     "SVM": SVC()
21 }
22 for name, model in models.items():
23     model.fit(X_train, y_train)
24     y_pred = model.predict(X_test)
25     accuracy = accuracy_score(y_test, y_pred)
26     print(name, "Accuracy:", accuracy)
```

Run exp 16

```
/Users/moshika/PycharmProjects/PythonProject/.venv/bin/python /Users/moshika/PycharmProjects/PythonProject/exp_16.py
KNN Accuracy: 1.0
Decision Tree Accuracy: 1.0
Naive Bayes Accuracy: 1.0
Logistic Regression Accuracy: 1.0
SVM Accuracy: 1.0
Process finished with exit code 0
```

Result: Different classification algorithms were successfully implemented and compared based on their accuracy.